

LAVIFiT: LATENT-ACTION-GUIDED VISION FINE-TUNING FOR SURGICAL INTERACTION RECOGNITION

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ABSTRACT

Understanding instrument-tissue interactions is essential for context-aware surgical AI and autonomous robotic surgery. Pretrained vision-language models (VLMs) and vision encoders offer an alternative to conventional interaction classifiers by transferring broad visual and semantic knowledge. However, adapting them to fine-grained surgical interactions remains challenging: (1) freezing the vision encoder depends entirely on pretrained representations that may retain noise and provide weak spatial localization, while (2) full fine-tuning can improve global semantic alignment without ensuring the encoder learns meaningful features in the correct Instrument-tissues interaction region. We address these limitations by introducing LAViFiT, an end-to-end latent-action-guided framework for vision-language fine-tuning: an inverse dynamics model captures the visual changes induced by each action, while a forward world model drives the encoder to represent action-relevant regions. A *Patch-level SIG Regularizer* further prevents local feature collapse without additional supervision, such as bounding boxes or pseudo-labels. Experiments across multiple encoders and datasets improve recognition and image-text alignment, while representation analyses show stronger grounding over the Instrument-tissues interaction region and more spatially coherent features. The source code is publicly available at: <https://marginlab.github.io/AI-for-healthcare/lavifit/>

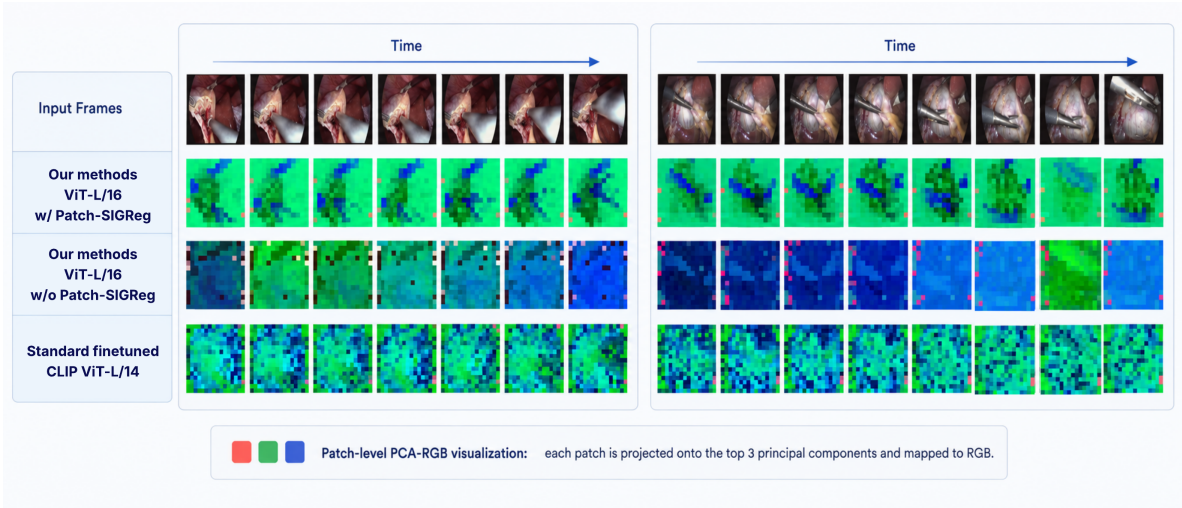


Figure 1: Visualization of vision encoder behavior under standard fine-tuning and our method, and the changes induced by the patch-level SIG regularizer.

1 Introduction

Robotic surgery has been widely adopted for its improved ergonomics, clinical outcomes, and reduced complication rates [42]. Yet current systems remain fully surgeon-operated, and surgical quality still depends heavily on surgeons’ experience and variability, motivating growing interest in surgical automation to achieve more consistent, reproducible performance. A prerequisite for such automation is recognizing *instrument-tissue interactions*: robots must understand what a surgeon is doing before they can assist or, eventually, learn skills from expert demonstrations. Instrument-tissue interaction recognition is commonly formalized as surgical action-triplet recognition, where the goal is to predict $\langle \text{instrument, verb, target} \rangle$ tuples [43]. Most triplet classification methods use three separate classifier heads for instrument, verb, and target prediction to learn discriminative decision boundaries [44, 46, 45]. Although these approaches have achieved substantial progress, they are still typically trained and evaluated under a closed-set label setting. As a result, they have limited generalization capability.

Recent progress in contrastive vision-language models (VLMs) [13, 40, 41, 39] has moved surgical AI toward a multimodal setting in which visual inputs are aligned with natural-language descriptions. Unlike conventional closed-set classifiers trained only with task-specific categorical labels, pretrained VLMs learn semantically aligned visual-language representations, while large-scale visual encoders provide broadly transferable visual features. Their exposure to diverse objects, actions, and visual contexts offers a strong initialization for adaptation to downstream tasks across different domains. Moreover, representing surgical action triplets as textual prompts rather than independent class IDs allows the model to exploit the compositional structure of $\langle \text{instrument, verb, target} \rangle$ labels and share information across related triplets, potentially improving generalization in action recognition [8, 32, 34]. Accordingly, recent surgical studies have adapted VLMs or visual encoders that pretrained on large-scale general-domain or surgical image/video data to downstream tasks such as action and phase recognition [36, 38, 63, 35]. However, standard adaptation strategies have two important limitations.

(1) Limited adaptation with frozen visual features. Freezing the visual encoder and tuning only the text branch can be ineffective when the pretrained visual features are noisy or spatially imprecise [31, 18, 20, 24]. This limitation is particularly relevant to surgical encoders pretrained from online videos with weakly aligned audio transcripts [40, 63, 41, 39, 37]. Similarly, masked self-supervised methods [7, 35, 36] learn by reconstructing randomly masked content but do not explicitly require the encoder to prioritize interaction-relevant motion or suppress task-irrelevant temporal variation. Consequently, irrelevant motion and background information may remain embedded in the frozen visual representations, limiting the effectiveness of frozen vision encoder fine-tuning.

(2) Limited spatial precision under full fine-tuning. Full image or video fine-tuning primarily optimizes global semantic alignment and does not guarantee that the resulting features preserve the localization, geometry, and fine-grained structure of task-relevant visual cues. The model may therefore continue to rely on shortcut features, including background context and irrelevant motion [17, 28, 27, 23, 25]. Even encoders that produce dense patch representations through self-distillation [14, 12] may retain artifacts or irrelevant information in their features [19].

To address the vision representation quality issues, existing approaches have primarily addressed representation-quality limitations through additional supervision. In the general domain, some methods align detector-derived object regions with text tokens [82, 83], while others use soft region grouping for region-text contrastive learning without explicit localization [8, 25]. Recent work further automates this process using segmentation-derived regions and pretrained contrastive models for object-sentence matching [27]. In the surgical domain, prior methods explicitly incorporate instrument trajectories or instrument-tissue pseudo-labels as additional supervision [49, 51, 53]. However, these approaches often require preprocessing to extract instrument appearance, motion, or region information. In the surgical domain, they also tend to emphasize instrument regions while overlooking the tissue being manipulated. These limitations motivate methods that learn action-relevant regions directly from temporal observations without manually constructed region labels or preprocessing pipelines.

Recent latent action model methods [75, 71] offer a promising way to learn action-sensitive representations without action labels or hand-crafted motion preprocessing. Given consecutive states, an inverse dynamics model (IDM) infers a compact latent action representation, while a forward world model (FWM) predicts the next state from the current state and that action. By restricting the capacity of the latent action, the IDM cannot simply transmit the entire future observation [71, 2]. LeWorldModel [80] further shows that end-to-end world-model training can encourage the visual encoder to capture meaningful motion properties, such as position and velocity, and theoretical study [47] shows IDM has ability capture the transition that caused by the agent, this suggests that jointly optimizing the encoder with world model next-state prediction objectives and combine with IDM may help overcome the limitations of conventional fine-tuning on downstream tasks that require detailed spatiotemporal representations, such as surgical action recognition. However, prior work has mainly focused on action discovery and control, and several methods in this line of research build on fixed pretrained visual backbones [21, 22, 71]. Consequently, the potential of end-to-end latent-action training

to reshape vision encoders for fine-grained VLM-based action recognition remains underexplored, particularly in terms of robustness to task-irrelevant variation, spatial coherence, and Instrument-tissues interaction region grounding.

Inspired by this direction, we introduce LAViFiT, an end-to-end latent-action-guided vision fine-tuning framework for surgical interaction recognition. It comprises a vision encoder, an inverse dynamics model (IDM), a forward world model (FWM), a clip aggregator, and a text encoder, all trained jointly. Unlike conventional IDMs that jointly process two consecutive states, our IDM operates on their latent difference, following Delta-JEPA [55], to discourage transition collapse and limit shortcuts when consecutive states are highly similar. We then compresses this difference through a low-dimensional bottleneck into a compact latent action with limited capacity for irrelevant motion noise. This latent action guides the FWM in predicting the next-frame representation from the current state, while the resulting prediction objective guides the vision encoder to retain transition information useful for forward prediction and surgical interaction recognition. The aggregator fuses the states and latent actions into a clip representation aligned with triplet text embeddings. We also introduce a *patch-level SIG regularizer* to prevent local feature collapse during end-to-end training, as illustrated in Fig. 1. Experiments show that this latent-action guidance improves recognition, produces more spatially coherent patch features, and better grounds frame-to-frame feature changes in instrument-tissue interaction regions.

In summary, our contributions are as follows:

- We introduce LAViFiT, a latent-action-guided fine-tuning framework that reshapes pretrained vision encoders toward interaction-relevant spatial and transition features. By jointly training an inverse dynamics model and a forward world model, LAViFiT adapts the encoder without region-level supervision or handcrafted motion preprocessing.
- We introduce a *patch-level SIG regularizer* that further prevents feature collapse during end-to-end training while keep computation cost tractable.
- We provide a systematic study of how LAViFiT fine-tuning with *patch-level SIG regularizer* reshapes vision encoder behavior, showing more spatially coherent features and stronger grounding of transition information in instrument-tissue interaction regions, together with consistent gains in recognition and image-text alignment across encoders and datasets.

2 Related Work

2.1 Surgical Action Triplet Recognition

The surgical action triplet (instrument, verb, target) was introduced by [1] as a fine-grained formalism for surgical workflow modeling. A triplet is fundamentally an *interaction*: it is defined not by static appearance but by a change of state and the agent that causes it. Nwoye et al. [4] first recognized triplets directly from video with Tripnet (CholecT40), later extended into the Rendezvous (RDV) attention model and the CholecT50 dataset [5], with standardized splits and metrics fixed in [48]. Subsequent methods improve temporal modeling (RiT [56], MT-FiST [57]), multi-task distillation [58], graph-based component dependencies [59], and parameter efficiency [60]. However, these classifier-based approaches rely heavily on well-annotated datasets and optimize closed-set category predictions, providing limited semantic alignment and transferability compared with pretrained vision encoders and VLMs.

2.2 Surgical Visual Pretraining and Downstream Adaptation

To develop strong foundation models for surgical tasks, recent work has explored large-scale pretraining of surgical vision encoders and vision-language models. Surgical VLMs commonly collect videos and narration from online surgical lectures and use automatically extracted video-text pairs for pretraining [40, 41, 39, 63]. However, such supervision can be noisy and weakly aligned, as the narration may not precisely describe the visual content or the fine-grained instrument-tissue interaction occurring at each moment, raising explainability concerns that are particularly important in medical applications [3]. In parallel, surgical vision encoders have been pretrained using masked image or video modeling, where masked regions are reconstructed in pixel space [36, 35], as well as JEPA-style objectives that predict latent representations rather than raw pixels [7]. Other work, such as SurgeNetXL [12], adopts DINO-style self-distillation to learn detailed surgical visual features. However, the representation quality produced by those pretrained vision encoders is still under-explored. Additionally, for downstream tasks such as action recognition, these pretrained encoders are typically adapted by either freezing the backbone and training only a classifier or text branch, or by fine-tuning the full vision encoder. However, frozen features may not sufficiently capture task-specific interactions, while conventional full fine-tuning can emphasize global semantic discrimination at the expense of fine-grained local structure [31, 18, 17, 28]. In this work, we propose *latent action modeling* methods, a simple fine-tuning framework that encourages the vision encoder to focus on action-relevant regions while preserving visual-semantic alignment with language.

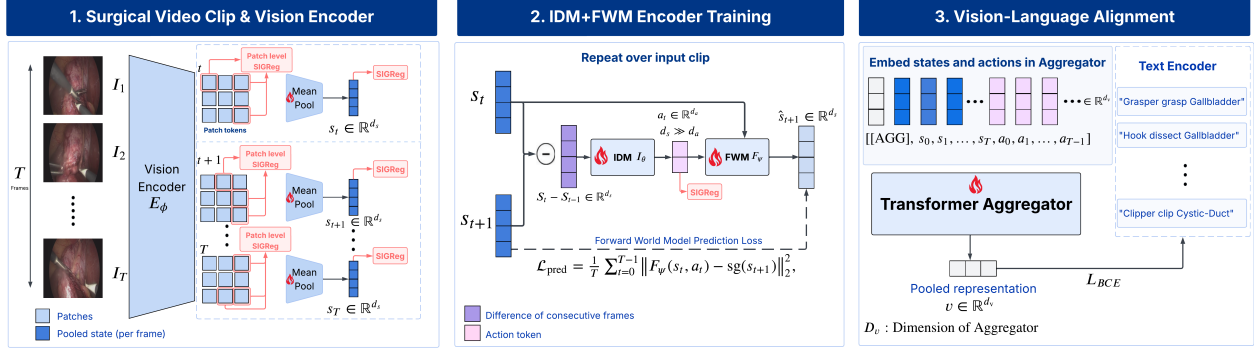


Figure 2: Overview of LAViFiT. The vision encoder maps a surgical clip to state tokens, while the IDM compresses consecutive-state changes into latent actions and the FWM predicts the next state using a stop-gradient target. Frame- and patch-level SIGReg prevent collapse, and the state-action sequence is aggregated and the aggregation token aligned with instrument-verb-target text embeddings.

2.3 Inverse Dynamics and Latent World Models

Latent action models infer the hidden cause of an observed transition without action labels by combining an inverse dynamics model (IDM), which encodes the change between consecutive states into a latent action, with a forward world model (FWM), which predicts the next state from the current state and the inferred action. This line of work has been studied mainly for robot control and policy learning [70, 2, 75, 71, 21, 22]. Early methods represented each transition using a discrete latent action selected from a finite set, as in ILPO [70]. LAPO [71] learned this discrete action space through vector quantization, mapping each continuous transition embedding to its nearest codebook entry, and Genie [72] later scaled this formulation to internet-scale video. Subsequent work explored embedding-space prediction with frozen visual encoders [2], robustness to visual distractors [73], and continuous latent actions for fine-grained policy learning [74]. More recently, LeWM [80] enabled end-to-end joint-embedding world-model training with SIGReg, showing that next-state prediction can encourage the encoder to capture physical properties such as position and velocity. On the other hand, recent theoretical analysis [47] showed that a compact IDM bottleneck can recover action-induced transitions without action labels, although structured nuisance variation may also be retained. However, these studies mainly target action discovery and robot control, and several methods mainly rely on frozen visual backbones [21, 22, 71]. In this work, we propose the LAViFiT method, which combines compact IDM learning with end-to-end world-model and vision-language fine-tuning objectives, and analyze how this joint training reshapes the behavior of pretrained vision encoders on a VLM-fine-grained detailed action recognition task.

3 Method

In surgical action-triplet recognition, given a video clip $\mathbf{X} = \{x_0, \dots, x_T\}$, we aim to predict the text of the $\langle \text{instrument, verb, target} \rangle$ present. LAViFiT formulate the video with *latent-action modeling*, inspired by [71]. We split the clip into frames and treat each frame as a state s_t . An inverse dynamics model (IDM) infers a compact latent action $a_t = I_\theta(s_t, s_{t+1})$ between consecutive states, and a forward world model (FWM) predicts the next state $\hat{s}_{t+1} = F_\psi(s_t, a_t)$ from the current state and this action. The pipeline is illustrated as Fig 2, we detail the components of LAViFiT in Section 3.1 and the loss design in Section 3.2.

3.1 Components

Given a clip of $T+1$ frames $(x_0, \dots, x_T)$, our model comprises a vision encoder E_ϕ , an inverse dynamics model (IDM) I_θ , a forward world model (FWM) F_ψ , a clip aggregator A_ω , and a text encoder G_χ . We explain each component in detail below.

Vision Encoder. For each frame, E_ϕ produces N patch tokens $\mathbf{p}_t = (p_{t,1}, \dots, p_{t,N})$ with $p_{t,k} \in \mathbb{R}^{d_s}$, and a pooled state

$$s_t = \text{pool}(\mathbf{p}_t) \in \mathbb{R}^{d_s}, \quad t = 0, \dots, T, \quad (1)$$

where $\text{pool}(\cdot)$ is a mean over patch tokens. For image backbones each frame is encoded independently; for video backbones, the clip is tiled into non-overlapping windows and s_t is the pooled state of window t , so E_ϕ yields a sequence of $T+1$ states either way. The patch tokens $\mathbf{p}_t$ are retained for the patch-level regularizer of Section 3.2.

Inverse dynamics model. Instead of using traditional IDM that take two consecutive states representation together as input, we use a Delta-IDM I_θ operates on the difference between consecutive states [55], $\delta_t = s_{t+1} - s_t$, this discouraging transition collapse and endpoint-state shortcuts that could bypass the latent-action bottleneck during forward prediction, and compresses it into a low-dimensional action token $a_t = I_\theta(\delta_t) \in \mathbb{R}^{d_a}$ for $t = 0, \dots, T-1$. It embeds δ_t to a hidden width d_m , applies an L_I -layer causal Transformer over the difference sequence, and linearly projects to the d_a -dimensional action space. This final projection is the *compression bottleneck*: since $d_a \ll d_s$, a_t can retain only the salient transition and prevent FWM from simply copying s_{t+1} .

Forward world model. The FWM F_ψ predicts the next state from the current state s_t and the latent action a_t , i.e. $\hat{s}_{t+1} = F_\psi(s_t, a_t)$, trained to match a stop-gradient target $\text{sg}(s_{t+1})$. It is a DiT-style Transformer with adaLN-zero conditioning [50]: the state s_t passes through the network, while the action a_t conditions each block via adaptive layer normalization, so a_t is the only path that carries information about the transition. Because that path is low-dimensional, driving Reconstructing s_{t+1} with F_ψ encourages the encoder to represent inter-state changes compactly while suppressing irrelevant noise.

Aggregator. A_ω is a Transformer that fuses the two streams into a single clip embedding. We project the state and compressed action token to the aggregator dimension d_v , then concatenate the $T+1$ states followed by the T actions into one token sequence, project each to a shared width d_ω , prepend a learnable aggregation token [AGG], and process the sequence with an L_ω -layer bidirectional Transformer:

$$v = A_\omega([\text{AGG}], s_0, s_1, \dots, s_T, a_0, a_1, \dots, a_{T-1}) \in \mathbb{R}^{d_t}, \quad (2)$$

where the clip embedding v is read from the output [AGG] token and linearly projected to the text embedding dimension d_t . Feeding both states (*what is present*) and actions (*what is happening*) lets the classifier condition triplet prediction on the inferred dynamics, not appearance alone.

Text encoder and prediction. We use EmbeddingGemma [11] as the text encoder G_χ . For each candidate triplet class $c \in \{1, \dots, C\}$, the encoder maps its textual description, e.g., “grasper retract gallbladder,” to token embeddings, which are mean-pooled to obtain a text representation $g_c \in \mathbb{R}^{d_t}$. We compute the classification logit for class c as

$$z_c = \tau \frac{\langle v, g_c \rangle}{\|v\|_2 \|g_c\|_2} + b, \quad (3)$$

where τ and b are learnable scale and bias parameters. The probability that triplet c is present is then given by $\hat{y}_c = \sigma(z_c)$.

3.2 Loss Design

The model is trained jointly using the recognition objective, the forward prediction objective, and the SIG regularizers. We use $\text{sg}(\cdot)$ to denote the stop-gradient operator.

Triplet and component alignment. The primary training signal is a multi-label classification loss that supervises both the complete triplets and their instrument, verb, and target components [34, 8]:

$$\mathcal{L}_{\text{cls}} = \lambda_{\text{ivt}} \text{BCEWithLogits}(z, y) + \sum_{q \in \{i, v, t\}} \lambda_q \text{BCEWithLogits}(z^q, y^q), \quad (4)$$

where $y \in \{0, 1\}^C$ is the multi-hot triplet label and $z = (z_1, \dots, z_C)$ contains the cosine-similarity logits for the C triplet prompts. The component labels y^i , y^v , and y^t are derived from the triplet annotations through fixed triplet-to-component mappings. Similarly, z^i , z^v , and z^t are obtained by comparing the clip representation v with the corresponding instrument, verb, and target text embeddings. For all experiments we use $\lambda_q = 1.0$.

Forward prediction. Classification alone shapes E_ϕ only toward static appearance and leaves the action a_t unconstrained—it need not explain any transition. We therefore add a forward-prediction loss,

$$\mathcal{L}_{\text{pred}} = \frac{1}{T} \sum_{t=0}^{T-1} \|F_\psi(s_t, a_t) - \text{sg}(s_{t+1})\|_2^2, \quad (5)$$

which forces (s_t, a_t) to jointly explain s_{t+1} , so a_t must carry exactly what distinguishes s_{t+1} from s_t . The stop-gradient $\text{sg}(\cdot)$ on the target prevents E_ϕ from trivially minimizing the loss by collapsing its states.

State and Action SIG Regularization. For end-to-end latent world model training, we adopt the SIGReg regularization used in LeWorldModel [80] and apply it to both the latent action and state representations. SIGReg matches the empirical distribution of the learned representations to a standard isotropic Gaussian using a sliced Cramér-Wold Epps-Pulley test [33]. Since a distribution in $\mathbb{R}^D$ is determined by all of its one-dimensional projections, we draw M random directions $u_1, \dots, u_M$ from the unit sphere $\mathbb{S}^{D-1}$, resampled at each step, and test Gaussianity on each projection instead of in $\mathbb{R}^D$ directly. For a set of latents $Z = \{z_i\}_{i=1}^N$,

$$\text{SIGReg}(Z) = \frac{1}{M} \sum_{m=1}^M T\left(\{\langle z_i, u_m \rangle\}_{i=1}^N\right), \quad (6)$$

where $T(\cdot)$ is the closed-form Epps-Pulley statistic, a non-negative measure of how far a set of scalars is from $\mathcal{N}(0, 1)$ that vanishes only when the two distributions match. We apply it to both latent streams:

$$\mathcal{L}_{\text{SIG}} = \text{SIGReg}\left(\{a_t\}_{t=0}^{T-1}\right) + \text{SIGReg}\left(\{s_t\}_{t=0}^T\right), \quad (7)$$

which encourages $a_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_a})$ and $s_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_s})$, so that each feature dimension carries unit variance while correlations between dimensions are suppressed, preventing representational collapse. Here $a_t \in \mathbb{R}^{d_a}$ and $s_t \in \mathbb{R}^{d_s}$ are the latent action and state tokens, and we use $M = 1024$ projections.

Per-patch Sig-Regularizer. The state-level regularizer can be satisfied even when most patch tokens collapse toward a single direction. Because instrument motion is often the most salient source of change in surgical videos, the model may preserve this dominant motion while discarding fine-grained tissue structure, as Figure 1 shows, which is also important for action-triplet recognition. We therefore additionally regularize the patch-level tokens. At each step we draw a random subset $\mathcal{K}_t \subset \{1, \dots, N\}$ of $|\mathcal{K}_t| = 32$ patches per frame to keep the cost tractable, and apply the same regularizer to the pooled set of sampled patch tokens:

$$\mathcal{L}_{\text{SIG}}^{\text{patch}} = \text{SIGReg}\left(\{\mathbf{p}_{t,k} : t=0, \dots, T, k \in \mathcal{K}_t\}\right), \quad (8)$$

which encourages $\mathbf{p}_{t,k} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_s})$. We use $M = 256$ projections for this term. The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{pred}} \mathcal{L}_{\text{pred}} + \lambda_s \mathcal{L}_{\text{SIG}} + \lambda_p \mathcal{L}_{\text{SIG}}^{\text{patch}}. \quad (9)$$

4 Experiment Design

We evaluate our method from three angles. **(1) Performance :** We first show whether latent-action supervision improves triplet recognition by comparing each encoder fine-tuned in the standard finetuning against the same encoder trained inside our method. **(2) Encoder behavior:** we then ask *how* the encoder’s representation changes, inspecting its features directly in section 4.2. **(3) IDM action-dimension compression analysis:** finally, we ask *what drives* the encoder behavior change in section 4.3. Our hypothesis builds on a known property of latent-action models [71]: the action dimension is kept small so the forward model cannot cheat by copying the next state through a high-capacity action. We find this same compression has another unremarked effect, which can force the Inverse Dynamic Model to encode the meaningful frame-to-frame change. We test this hypothesis with three experiments in section 4.3.

Datasets and encoders. We evaluate on two surgical action-triplet benchmarks. **CholecT50** [43] comprises 50 laparoscopic cholecystectomy videos (90,489 frames at 1 fps) with 100 triplet classes over 6 instruments, 10 verbs, and 15 targets. **ProstaTD** [10] comprises 21 robot-assisted radical prostatectomy videos ($\sim 72\text{K}$ frames) with 89 triplet classes over 7 instruments, 10 verbs, and 10 targets. We use each dataset’s official split: The standard CholecT50 RDV split and ProstaTD’s 5-fold cross-validation, and report the triplet mAP (mAP_{ivt}) and per-component APs (mAP_I , mAP_V , mAP_T) using *ivtmetrics* [43]. We benchmark encoders from two pretraining regimes: *general-domain*—CLIP ViT-L/14 [13], DINOv2-L/14 [14], V-JEPA 2/2.1 [78, 15], and TimeSformer [9]; and *surgical-domain*—EndoViT [36], SurgeNet [12], and the surgical VLMs SurgVLP [39], HecVLP [40], PeskaVLP [41], and LEMON [38]. We instantiate our method on three image encoders chosen to represent distinct pretraining paradigms: **ViT-L/16**, a vanilla vision transformer serving as the most basic encoder; **DINOv2-L/14**, a self-distillation encoder representative of methods that produce dense patch features (general-domain); and **SurgeNet-L/14**, which build on top of DINOv2’s architecture but is pretrained on surgical data, this forming a controlled pair with DINOv2 that isolates general versus surgical pretraining of the same backbone. For all experiments, the video-encoder baselines and LAViFiT construct an eight-frame clip from each labeled frame and its seven preceding frames, assigning the clip the label of that frame. Image-encoder baselines instead process each labeled frame independently.

Instrument-tissues interaction region mask. We first define the Instrument-tissues region used in the experiments of Sec. 4.2 and Sec. 4.3. For each frame x_t , we obtain an Instrument-tissues interaction region mask with the segmentation-estimation method of SASVI [30], which propagates the sparse CholecSeg8k [29] annotations into dense per-frame masks for the CholecT50 [43] test videos. We retain the eight CholecSeg8k classes that map to CholecT50 triplet instruments and targets: two instrument classes (*grasper*, *hook*) and six tissue/target classes (*abdominal wall*, *liver*, *blood/fluid*, *cystic duct*, *gallbladder*, *hepatic vein*). We define the Instrument-tissues interaction region as following.

$$y_t = (i_t, v_t, r_t), \quad (10)$$

where i_t is the instrument and r_t is the target tissue/anatomy, the Instrument-tissues interaction region mask is defined as

$$M_t^{\text{act}} = M_t^{\text{inst}}(i_t) \cup M_t^{\text{tgt}}(r_t). \quad (11)$$

Thus, M_t^{act} contains only the instrument and target regions contained in the ground truth triplet, Other area out side of Instrument-tissues interaction region are consider as background mask. Applying this definition to the test split yields 12k masked test consecutive frames for the experiments in section. 4.2 and section. 4.3.

4.1 Recognition performance

We report triplet recognition-mAP_{IVT} with the instrument/verb/target component mAPs—on CholecT50 and ProstaTD (5-fold). **Standard fine-tuning.** We first fine-tune a range of image and video encoders directly for classification, aligned to the text head, where image encoders use single frames, video encoders on the whole clip. This conventional per-frame / per-clip baseline is reported for all encoders in our study. **Our LAViFiT fine-tuning method.** Among these encoders, we select a representative subset and train them within our method. **Ablations.** We further train two reduced settings. First, a *states-only* model that removes the IDM/FWM: the aggregator A_ω temporally fuses the per-frame state tokens. This is basically a CLIP4Clip-style [6] model, this isolates the contribution of the latent-action module. Second, we ablate the patch-level SIGReg, training our method with and without it to measure its effect. For a fair comparison, each encoder uses the same EmbeddingGemma text encoder [11] and triplet alignment loss described in 3.2, and same vision and text encoder learning rates in both the baseline and our method, ensuring that any performance gain is attributable to the latent-action module rather than differences in hyperparameter tuning. All recognition results are reported using 128-dimensional ($d_a = 128$) action tokens. Additional implementation details are provided in section A.1.

4.2 Encoder behavior

Calinski-Harabasz patch cluster quality. We measure how well the encoder separates individual surgical objects in its feature space, using the Instrument-tissues interaction region mask M_t^{act} defined above, which is obtained with the SASVI [30] segmentation-estimation method that propagates the sparse CholecSeg8k [29] annotations into dense per-frame masks. We shrink this mask down to the patch grid, giving each patch the label that covers most of its pixels. Each object class inside M_t^{act} then forms one cluster, and all patches outside it form a single background cluster. We only score frames where a ground-truth triplet has both its instrument and its target visible in the mask, which gives the 12k frames noted above. With K clusters and N retained patches, the Calinski-Harabasz score [26] is the ratio of between- to within-cluster dispersion,

$$\text{CH} = \frac{\sum_{k=0}^{K-1} n_k \|\bar{\mathbf{p}}_k - \bar{\mathbf{p}}\|_2^2}{\sum_{k=0}^{K-1} \sum_{\mathbf{p}_n \in \mathcal{C}_k} \|\mathbf{p}_n - \bar{\mathbf{p}}_k\|_2^2} \cdot \frac{N - K}{K - 1}, \quad (12)$$

where n_k is the size of cluster $\mathcal{C}_k$, $\bar{\mathbf{p}}_k$ its mean, and $\bar{\mathbf{p}}$ the mean over all retained patches. Higher CH means tighter, better-separated objects. To relate separability to recognition, we z -normalize each model’s per-frame CH, pool the compared models into five equal-count bins, and report mean CH and mAP per bin.

4.3 IDM dimension compression analysis

The inverse model compresses the inter-frame transition $\delta_t = s_{t+1} - s_t$ into a low-dimensional latent action, which the forward model uses to predict the next state. Prior work primarily introduces this capacity constraint to prevent the inverse model from trivially transmitting the future state to the forward model [71, 2]. We further investigate how this constraint reshapes the visual encoder itself under the VLM training setting. Specifically, We hypothesize that joint IDM/FWM training uses the compressed latent-action pathway to guide vision encoder adaptation. Because the FWM must predict the next state from only the current state and a limited-capacity latent action, this constraint encourages the encoder to produce compact features whose transitions are grounded in the Instrument-tissues interaction region. We evaluate this hypothesis through three complementary experiments.

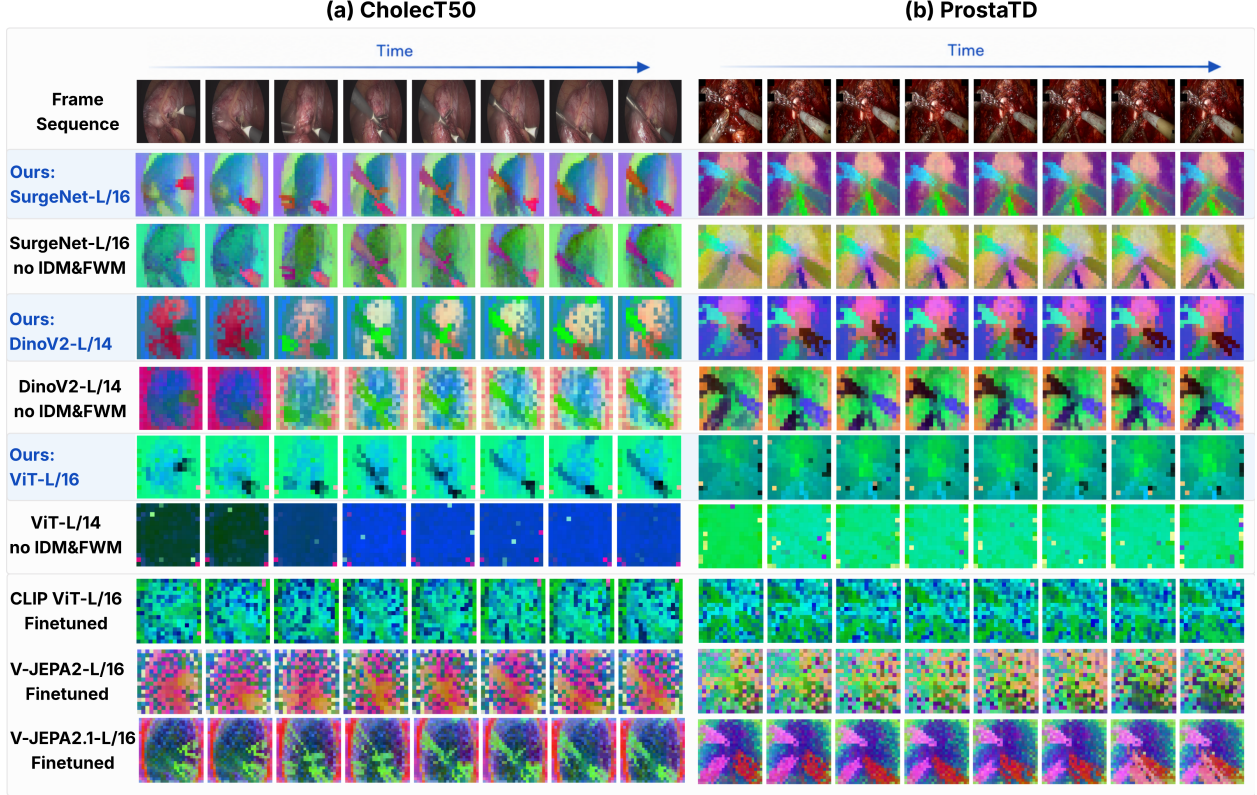


Figure 3: PCA-RGB visualizations of encoders trained with and without our latent action framework on (a) CholecT50 [43] and (b) ProstaTD [10]. All variants of our method are trained with IDM, FWM, and patch-level SIGReg. V-JEPA2 and V-JEPA2.1 figures are visualized using tablet tokens.

(1) Latent-action bottleneck compression. We test whether the bottleneck compresses the encoder’s representation. For each encoder we take consecutive-frame state differences $\delta_t = s_{t+1} - s_t$ over the test split and form their covariance $\text{Cov}(\delta)$, with eigenvalues $\lambda_1 \geq \dots \geq \lambda_D$ and $p_i = \lambda_i / \sum_j \lambda_j$. We report three scale-invariant measures of its effective dimensionality: the *participation ratio* $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ and the *entropy rank* $\exp(-\sum_i p_i \log p_i)$ (two estimates of the effective number of active directions), and d_{90} , the number of leading components capturing 90% of the variance. Lower values for the full model than the no-IDM/FWM baseline mean the change is compressed into fewer directions.

(2) Change-subspace grounding. We then test *where* compressed representation directions lie. We take the top- r eigenvectors U_r of $\text{Cov}(\delta)$ ($r = 128$) the dominant directions of frame-to-frame change. On each frame, project every patch’s temporal change onto them, where ϕ_p^t denotes the feature of patch p at frame t and $\Delta\phi_p = \phi_p^t - \phi_p^{t-1}$ measures how that patch changes between consecutive frames $t-1$ and t . We score each patch use its magnitude-invariant fraction $\text{norm} = \|U_r^\top \Delta\phi_p\|^2 / \|\Delta\phi_p\|^2$ (alignment only). We threshold the per-patch map (top 25%) and compute IoU against the SASVI [30] generated Instrument-tissues interaction region mask (instrument $\cup$ target tissue) (11). A higher IoU means the dominant change directions land on the Instrument-tissues interaction region. We evaluate on three image encoders ViT-L, DINOv2-L, and SurgeNet-L, both the baseline without IDM/FWM and our full method, and additionally report baseline results for three video encoders—TimeSformer, V-JEPA 2, and V-JEPA 2.1. For each video encoder, we mean-pool the frame-level or tubelet-level tokens to obtain “per-state” representations and then use them as s_t and s_{t+1} to perform the same experiments described above.

(3) Action-dimension ablation. Finally, we vary the action dimension d_a and, for each, re-measure the above two experiments and their average precision to show how dimension compression affects the model’s overall performance, CH score and Subspace Grounding IoU score.

(a) Main comparison on CholecT50 and ProstaTD										
Method	CholecT50(RDV)				ProstaTD (5-FOLD)					
	mAP _{IVT}	mAP _I	mAP _V	mAP _T	mAP _{IVT}	mAP _I	mAP _V	mAP _T		
<i>General-domain pretraining</i>										
CLIP ViT-L/14 [†]	13.85	84.08	55.19	34.96	20.02±1.65	90.12±1.27	63.32±4.47	49.31±2.78		
DINOv2-L [†]	16.72	88.58	61.78	39.60	25.71±1.65	94.65±1.19	69.09±4.22	57.34±3.00		
V-JEPA 2 [‡]	17.01	90.61	63.28	39.94	25.02±1.68	92.39±1.37	69.49±4.29	53.24±4.37		
V-JEPA 2.1 [‡]	17.97	91.37	63.90	41.55	25.49±1.68	93.82±1.79	72.38±3.62	56.93±2.69		
VL-JEPA [‡] * / V-JEPA 2 backbone	14.49	84.78	57.98	34.92	20.75±1.20	85.23±1.84	63.22±5.10	50.09±2.31		
TimeSformer [‡]	14.29	84.34	57.49	36.80	18.29±1.54	84.23±1.87	61.22±4.31	48.69±2.89		
<i>Surgical-domain pretraining</i>										
EndoViT [†]	14.25	83.49	57.25	35.38	19.81±1.57	87.89±1.14	60.69±5.23	49.32±2.58		
SurgeNet, DINOv2 backbone [†]	18.04	90.23	63.40	41.83	25.32±1.53	94.19±0.91	70.66±3.99	58.84±2.77		
SurgVLP [†]	12.68	71.18	47.70	33.35	15.34±1.23	77.64±1.21	51.30±3.04	42.37±2.09		
HecVLP [†]	12.67	76.07	50.92	32.09	16.52±1.06	80.73±2.58	54.68±3.77	44.76±2.34		
PeskaVLP [†]	12.12	75.20	48.71	30.29	17.17±1.35	81.92±1.76	55.87±4.44	46.20±1.79		
LEMON [†]	15.86	85.32	60.26	39.35	23.14±1.12	91.17±0.51	65.01±4.58	54.34±2.76		
LAViFiT – ViT-L [†]	17.22	90.91	62.31	39.64	23.58±1.78	91.96±1.11	67.24±4.73	54.13±3.29		
LAViFiT – DINOv2-L [†]	18.03	93.03	63.60	41.16	26.28±1.64	95.38±1.54	70.59±5.20	58.20±3.09		
LAViFiT – SurgeNet-L [†]	18.50	92.97	65.34	42.34	27.51±1.28	96.17±1.28	72.52±3.68	59.06±2.41		
(b) Ablation study with ViT-L										
Configuration	SIG _p	IDM/FWM	CholecT50				ProstaTD			
			mAP _{IVT}	mAP _I	mAP _V	mAP _T	mAP _{IVT}	mAP _I	mAP _V	mAP _T
Full model	✓	✓	17.22	90.91	62.31	39.64	23.58±1.78	91.96±1.11	67.24±4.73	54.13±3.29
No patch-level SIGReg	×	✓	16.65	90.51	60.44	38.79	23.34±1.51	91.65±0.82	66.19±4.10	53.09±2.50
No patch-level SIGReg, IDM, or FWM	×	×	15.47	88.57	59.94	37.51	18.88±1.18	85.46±0.73	58.16±3.86	48.19±3.00
No action tokens at inference	✓	✓	15.79	86.60	59.41	38.55	21.84±1.11	90.14±1.02	63.14±2.43	51.62±3.00
(c) Ablation study with DINOv2-L										
Configuration	SIG _p	IDM/FWM	CholecT50				ProstaTD			
			mAP _{IVT}	mAP _I	mAP _V	mAP _T	mAP _{IVT}	mAP _I	mAP _V	mAP _T
Full model	✓	✓	18.03	93.03	63.60	41.16	26.28±1.64	95.38±1.54	70.59±5.20	58.20±3.09
No patch-level SIGReg	×	✓	17.54	91.57	62.68	40.01	25.64±1.14	94.13±1.70	69.86±3.25	57.58±3.65
No patch-level SIGReg, IDM, or FWM	×	×	16.28	89.85	60.88	39.71	23.61±1.62	92.75±1.78	66.98±3.71	55.24±3.44
No action tokens at inference	✓	✓	16.87	87.52	60.86	39.80	24.75±1.83	94.02±1.09	68.72±3.13	58.01±3.55
(d) Ablation study with SurgeNet-L										
Configuration	SIG _p	IDM/FWM	CholecT50				ProstaTD			
			mAP _{IVT}	mAP _I	mAP _V	mAP _T	mAP _{IVT}	mAP _I	mAP _V	mAP _T
Full model	✓	✓	18.50	92.97	65.34	42.34	27.51±1.28	96.17±1.28	72.52±3.68	59.06±2.41
No patch-level SIGReg	×	✓	18.36	92.06	64.04	41.63	27.12±1.45	95.72±1.78	71.88±4.76	59.29±3.09
No patch-level SIGReg, IDM, or FWM	×	×	18.11	92.45	63.81	42.29	26.28±1.19	95.58±1.15	70.98±3.38	57.48±2.32
No action tokens at inference	✓	✓	17.89	91.48	63.16	41.81	26.23±1.52	94.25±2.09	69.18±2.98	58.03±2.19

Table 1: Main results and ablations on CholecT50 and ProstaTD. We report instrument, verb, target, and triplet mAP. ProstaTD results are mean ± standard deviation across the 5 cross-validation folds; CholecT50 uses a single run. [†] and [‡] denote image and video encoders, and * denotes our reimplement of VL-JEPA. SIG_p is the patch-level SIG regularizer. Removing SIG_p and IDM/FWM yields a CLIP4Clip-style baseline [6]. The no-action-token variant is evaluated only at inference.

5 Experiment Results

5.1 Recognition Performance

Table 1 reports triplet recognition results on CholecT50 and ProstaTD. Across all three encoders, our latent-action method improves over the corresponding standard fine-tuned baseline in mAP_{IVT}. Our SurgeNet-L model achieves the best overall result on both datasets, reaching 18.50 and 27.51 on CholecT50 and ProstaTD, respectively, and outperforms all general- and surgical-domain baselines, including video encoders and surgical vision-language models.

The improvements are observed across the instrument, verb, and triplet metrics, indicating that the gains are not limited to object appearance alone. Compared with the fine-tuned CLIP ViT-L/16 baseline, LAViFiT with ViT-L/14 improves mAP_{IVT} by 3.37 points on CholecT50 and 3.56 points on ProstaTD. It also achieves results comparable

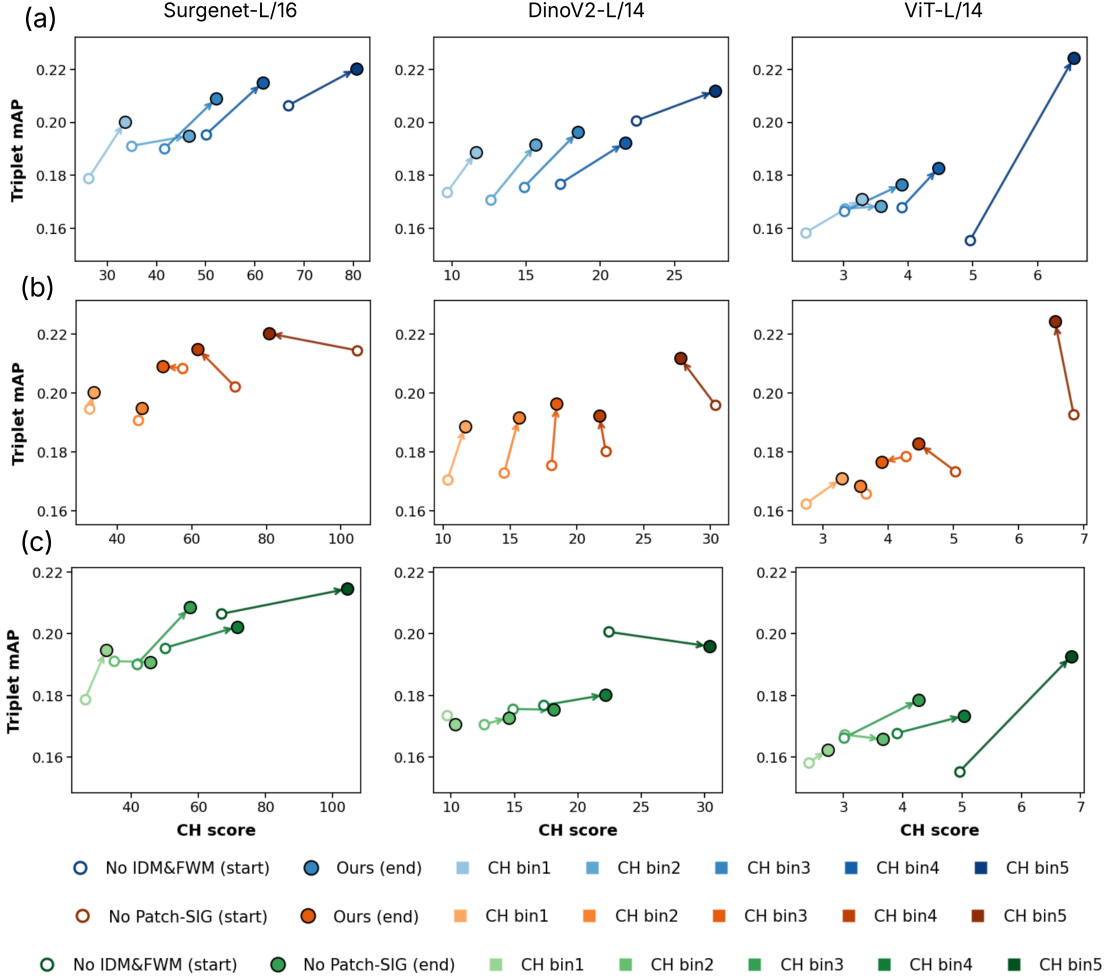


Figure 4: **Per-bin CH $\rightarrow$ mAP trajectories** ($d_a = 128$, triplet mAP). Test frames are split into five CH quintile bins (z -scored, pooled per pair); each arrow joins the mean CH (x) and triplet mAP_{VT} (y) of the two models (start hollow, end filled), bins colored dark $\rightarrow$ bright for low $\rightarrow$ high CH. CH is over the Instrument-tissues interaction region object classes plus one background class, so it rewards objects separating from each other *and* from background. Columns: ViT-L, DINOv2, SurgeNet. (a) baseline (no IDM/FWM) $\rightarrow$ LAViFiT (blue); (b) LAViFiT without Patch-SIG $\rightarrow$ LAViFiT (orange). Up-right motion = higher separability and recognition; (c) baseline (no IDM/FWM) $\rightarrow$ LAViFiT (green) without Patch-SIG. Up-right motion = higher separability and recognition

to those of the standard fine-tuned DINOv2-L/14 and V-JEPA 2 models. When our method is applied to the stronger surgically pretrained SurgeNet encoder, it achieves the best performance across multiple evaluation metrics.

Compared with V-JEPA 2 [79], LAViFiT with SurgeNet-L model improves $mAP_{IVT/I/V/T}$ from 17.01/90.61/63.28/39.94 to 18.50/92.97/65.34/42.34 on CholecT50 and from 25.02/92.39/69.49/53.24 to 27.51/96.17/72.52/59.06 on ProstaTD. Compared with V-JEPA 2.1 [15], an encoder designed for both dense feature extraction and video understanding, our method improves $mAP_{IVT/I/V/T}$ from 17.97/91.37/63.90/41.55 to 18.50/92.97/65.34/42.34 on CholecT50 and from 25.49/93.82/72.38/56.93 to 27.51/96.17/72.52/59.06 on ProstaTD, with particularly clear gains in instrument and target recognition.

Ablations. Parts (b)-(d) of Table 1 present ablations that isolate the contribution of each component. Removing patch-level SIGReg generally reduces performance, while additionally removing the IDM/FWM module causes a substantially larger drop. For example, on ProstaTD with ViT-L, mAP_{IVT} decreases from 23.58 for the full model to 23.34 without patch-level SIGReg and to 18.88 without patch-level SIGReg, IDM, or FWM. The joint benefit of IDM/FWM and patch-level SIGReg is larger for the weaker ViT-L encoder than for the more strongly pretrained DINOv2-L

Encoder	Setting	PR	Ent. rank	d_{90}	IoU _{norm} $\uparrow$
<i>Image encoders</i>					
LAViFiT-ViT-L	No IDM&FWM	156.7	410.1	497	0.057
	No Patch-level SIG	13.6	27.4	22	0.060
	With Patch-level SIG	34.4	88.1	164	0.074
LAViFiT-DINOv2-L	No IDM&FWM	123.0	409.2	595	0.081
	No Patch-level SIG	17.7	27.4	19	0.080
	With Patch-level SIG	26.0	77.1	185	0.137
LAViFiT-SurgeNet-L	No IDM&FWM	98.7	304.6	433	0.111
	No Patch-level SIG	16.5	55.9	125	0.096
	With Patch-level SIG	43.5	151.6	295	0.181
<i>Video encoder references</i>					
TimeSformer	Standard FT	-	-	-	0.060
V-JEPA 2	Standard FT	-	-	-	0.090
V-JEPA 2.1	Standard FT	-	-	-	0.149

Table 2: Mechanism analysis on CholecT50. For image encoders, No IDM&FWM denotes the CLIP4Clip-style temporal baseline, No Patch-level SIG retains IDM/FWM but removes the patch-level regularizer, and With Patch-level SIG denotes the full model. PR, entropy rank, and d_{90} are computed from $\delta_t = s_{t+1} - s_t$; lower values indicate more concentrated transition variance. Higher IoU_{norm} indicates stronger Instrument-tissues interaction region grounding. Video encoders are Standard-FT references, and dashes denote unreported transition statistics.

and SurgeNet-L encoders. On ProstaTD, the full framework improves mAP_{IVT} over the baseline without these components by 4.70 points for ViT-L, compared with 2.67 for DINOv2-L and 1.23 for SurgeNet-L. On CholecT50, the corresponding gains are 1.75, 1.75, and 0.39, respectively. This trend suggests that latent-action learning and patch-level regularization provide greater benefits when the pretrained encoder has weaker task-specific structure, while strongly pretrained dense encoders leave less room for improvement. Removing the action tokens only at inference also degrades performance: for ViT-L, mAP_{IVT} decreases from 17.22 to 15.79 on CholecT50 and from 23.58 to 21.84 on ProstaTD.

5.2 Encoder behavior

We next examine *how* the method reshapes the encoder. Figure 3 visualizes the top three principal components of the per-patch features as RGB. Compare with the standard finetuned V-JEPA2 and V-JEPA2.1 and CLIP-ViT-L, across all three encoders, our method produces more spatially coherent PCA-RGB maps, in which patches belonging to the same anatomical structure or instrument form more uniform color regions, while sharper color transitions separate object boundaries. It also suppresses scattered background noise, making instruments, tissues, and their interaction regions easier to distinguish than under standard fine-tuning.

To quantify these changes, we measure the Calinski-Harabasz (CH) separability of surgical objects within the Instrument-tissues interaction region. We sort all test frames by CH score and split them into five equal-size bins, from lowest to highest separability. Figure 4 (a) compares the baseline without IDM/FWM with LAViFiT full model within each CH-score bin. Each arrow starts from the baseline result and ends at the corresponding result of our method. Across all three encoders and the evaluated bins, most arrows shift toward the upper right, indicating simultaneous improvements in feature separability, measured by the CH score, and triplet-recognition performance, measured by mAP_{IVT} . Comparing the baseline with LAViFiT without the patch-level regularizer (Figure 4(c)), the IDM/FWM raises the CH score for every encoder and bin, yet mAP_{IVT} still falls in some bins—e.g., Bins 1, 3, and 5 for DINOv2-L/14. Thus, we then further compare LAViFiT with and without the *patch-level SIG regularizer* 4(b). For samples that already exhibit well-separated patch features without using *patch-level SIG regularizer*, the regularizer may reduce the CH score or provide only limited additional improvement. This is observed for Bins 3-5 with SurgeNet-L/16 and Bins 4-5 with DINOv2-L/16, whose self-distillation pretraining already produces spatially coherent dense features, as well as for Bins 4-5 with ViT-L/16. This behavior is expected because the regularizer encourages feature diversity to prevent local collapse rather than directly maximizing class separability. Moreover, for several bins, including Bins 3-5 for SurgeNet-L/16, Bins 4-5 for DINOv2-L/14, and Bins 2, 4, and 5 for ViT-L/14— mAP_{IVT} still increases even when the CH score decreases.

5.3 IDM analysis

Finally, Table 2 examines how IDM changes the encoder’s output representations and their Instrument-tissues interaction region grounding.

(a) Compression. The effective rank of the state-difference covariance $\text{Cov}(\delta)$ decreases substantially under our objective. The participation ratio drops from 156.7 to 34.4 for ViT-L, from 123.0 to 26.0 for DINOv2, and from 98.7 to 43.5 for SurgeNet, indicating that the vision encoder represents inter-state changes within a much lower-dimensional subspace. Without the patch-level SIG regularizer, However, the IDM-FWM objective compresses the representation far more aggressively, yielding PR values of 13.6, 17.7, and 16.5 and d_{90} as low as 19. This concentrates the features into a small number of directions, indicating a symptom of dimensional collapse—leaving little capacity to represent the diversity of visual transitions.

(b) Grounding. We back-project the per-patch changes onto the top- r eigenvectors of $\text{Cov}(\delta)$ and measure their IoU with the Instrument-tissues interaction region. The results show that the compressed transition subspace becomes more strongly aligned with instrument-tissue interactions: IoU_{norm} increases from 0.057 to 0.074 for ViT-L, from 0.081 to 0.137 for DINOv2, and from 0.111 to 0.181 for SurgeNet. Moreover, LAViFiT with the SurgeNet-L/16 encoder achieves a higher IoU than the fine-tuned TimeSformer, V-JEPA 2, and V-JEPA 2.1 video encoders. The *patch-level SIG regularizer* further improves grounding, increasing IoU_{norm} from 0.060 to 0.074, from 0.080 to 0.137, and from 0.096 to 0.181, respectively.

(c) Action-dimension ablation. We provide an additional ablation study of the action-token dimension in Appendix A.2.

5.4 Discussion

Based on the results in Sections 5.2 and 5.3, we interpret our findings through the following three questions.

Why does joint IDM-FWM training improve spatial separability (CH)? Because the action token is much lower-dimensional than the state, only a limited amount of transition information can pass through the latent-action bottleneck. Meanwhile, the FWM must use this compressed representation to predict the next state. This discourages the model from preserving small, irregular frame-to-frame variations and instead favors compact and repeatable changes that are useful for forward prediction. Together with the surgical interaction-recognition objective, this bias encourages the encoder to retain changes associated with instrument and tissue motion rather than spatially diffuse nuisance variation. Representing these structures distinctly makes their changes easier to encode and predict, leading to tighter within-object features and clearer separation between instruments, tissues, and background. This is consistent with the higher CH scores observed after adding the IDM and FWM. In contrast, standard appearance-based fine-tuning imposes no explicit transition bottleneck, allowing visually similar structures and irrelevant variations to remain entangled, as reflected by the less spatially coherent PCA-RGB maps in Figure 4 and Figure 3. To better shows how action dimension compression affect the CH score we also provide ablation study in Appendix A.2.

Why does the patch-level SIG regularizer reduce CH on samples that are already well separated? Because the regularizer enforces feature *diversity*, not class *separability*. It constrains every patch location to contribute to the feature distribution, which necessarily introduces additional variation *within* each region. On frames whose patch features are already cleanly clustered. For example, bins 3-5 of SurgeNet-L and Bins 4-5 of DINOv2-L, whose self-distillation pretraining already yields coherent dense features, this added within-class variation slightly lowers CH without a corresponding gain in between-class spread. The regularizer is an anti-collapse constraint rather than a separability objective: it is most useful where local collapse is the limiting factor. Consistently, mAP_{IVT} still improves in several of these bins even as CH decreases, indicating that the recognition gains of LAViFiT do not arise solely from higher CH, which lead to questions below.

Why does higher CH alone not explain the performance gain, and what does the patch-level SIG regularizer contribute? CH and IoU_{norm} measure complementary properties: CH describes how well object features separate *within a frame*, whereas IoU_{norm} measures where the dominant *state-to-state change* directions are spatially concentrated. Although the IDM bottleneck improves feature separability, it also concentrates transition variance into very few directions, reducing the effective rank of $\text{Cov}(\delta)$ by approximately 6-12 $\times$, depending on the encoder (Table 2). Under such aggressive compression, the remaining transition subspace can become dominated by generic or spatially diffuse motion rather than clearly separating Instrument-tissues interaction region changes from background variation. Consequently, many high-scoring patches fall outside the Instrument-tissues interaction region, resulting in low IoU_{norm} . The *patch-level SIG regularizer* restores part of the lost transition rank, allowing action-related and generic motion variations to be represented by more distinct feature directions. Accordingly, IoU_{norm} increases for every encoder. In summary, CH reflects *what visual structures the encoder can distinguish*, whereas IoU_{norm} reflects *where the encoder’s dominant transition directions are grounded*. The IDM/FWM primarily improves spatial separability, while the patch-level SIG regularizer improves grounding within the Instrument-tissues interaction region. Although the regularizer sometimes lowers CH, its anti-collapse effect restores the diversity of feature directions on top of the spatially coherent features produced by the IDM&FWM, and it is the combination of the two that yields the recognition gains.

6 Conclusion and Future Work

In this work, we introduce LAViFiT, a fine-tuning framework that formulates surgical instrument-tissue interaction recognition through latent-action modeling. An inverse dynamics model compresses the transition between consecutive encoder states into a low-dimensional action representation, while a forward world model predicts the next state from the current state and the inferred action. Combined with a patch-level distribution regularizer, LAViFiT consistently outperforms standard fine-tuning across both general-domain and surgical-domain image encoders on CholecT50 and ProstaTD. Our mechanism analysis further shows that compact IDM-FWM training concentrates feature transitions into fewer dominant directions and improves their spatial alignment with action-relevant instrument-tissue regions. Crucially, the patch-level SIG regularizer prevents this compression from collapsing too aggressively: by restoring per-patch feature diversity, it keeps the dominant change directions concentrated on the Instrument-tissues interaction region rather than diffuse background motion, which is what turns the compressed representation into stronger Instrument-tissues interaction region grounding. Future work will extend the framework to pretrained video encoders for longer-horizon video understanding and investigate its use for large-scale pretraining on web-scale surgical video datasets. Additionally, end-to-end world model style objective with SIG-style regularization is highly sensitive to hyperparameter choices, while exhaustive tuning is impractical. In this work, we primarily investigate the potential of latent-action modeling for VLM-based surgical interaction recognition. Future work should explore more efficient hyperparameter optimization and more robust regularizers that provide stable performance with less task-specific tuning.

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A Appendix

A.1 Implementation details.

All three encoders share the same latent-action components and differ only in the backbone. Given an 8-frame clip, the encoder produces one state per frame with $d_s=1024$. The **inverse dynamics model (IDM)** is a 2-layer causal Transformer ($d_m=768$, 8 heads) that operates on consecutive state differences $s_{t+1}-s_t$ and projects them through a $d_a=128$ bottleneck to the latent action. The **forward world model (FWM)** is a 6-layer DiT-style Transformer (16 heads) with adaptive layer normalization (adaLN-zero) that predicts the next state from the current state conditioned on the action, trained against a stop-gradient target. The **aggregator** is a 4-layer Transformer ($d=768$, 8 heads) over the interleaved state and action tokens, and the **text encoder** is EmbeddingGemma-300M. The forward-prediction, action-SIGReg, state-SIGReg, and per-patch-SIGReg losses are weighted by 1.0, 0.1, 0.1, and 0.3, respectively; for the per-patch SIGReg we randomly subsample 32 patches per frame to keep its $\mathcal{O}(N^2)$ cost tractable. We train for 10 epochs with a cosine schedule (1 warmup epoch), batch size 8, weight decay 0.01, and bf16 mixed precision. We summarize the per-component learning rates in Table 3; for a controlled and fair comparison, each encoder and its corresponding baseline use the same learning rates for the vision and text encoders. All experiments were conducted using four NVIDIA RTX PRO 6000 Blackwell GPUs.

A.2 Ablation study on dimension of action tokens

For all recognition benchmarks, we set the action dimension to 128 across methods to ensure a fair comparison. Figure 5 further sweeps the action bottleneck dimension d_a from 64 to 1024. The results show that different vision encoders favor different action dimensions for different components in action triplet, and setting the action dimension too small or equal to the state dimension often leads to lower recognition performance.

We provide additional experiments result under same setting as 4.3 to show how action token d_a change the subspace grounding and CH score. As shown in Figure 6, subspace-grounding IoU and CH score exhibit a similar pattern: both an overly small and an overly large action dimension d_a tend to reduce grounding IoU and cluster quality, with an intermediate d_a (around 256) giving the strongest values. A too-small bottleneck lacks the capacity to encode the action-relevant transition directions, while a too-large one dilutes them across excess dimensions, weakening the concentration of change onto the Instrument-tissues interaction region. For SurgeNet [12], its grounding IoU stays nearly flat across d_a (ranging only between 0.180 and 0.190). whereas its CH score still shows the intermediate- d_a peak. We attribute this to SurgeNet’s surgical-domain pretraining, its features are already strongly aligned with instrument/tissue regions, so the spatial localization of the dominant transition subspace is already near saturation and largely insensitive to the bottleneck size. CH, which measures how cleanly the feature space separates objects in the Instrument-tissues interaction regions rather than where they land, shows larger changes when the action-token dimension changes for SurgeNet. Its CH score goes from 61.35 at $d_a=256$ to 52.26 at $d_a=1024$, about a 15% drop. This shows that our method further improves the SurgeNet vision encoder on the feature-separation part.

Table 3: Per-component learning rates for each vision encoder. **Encoder** is the vision backbone’s rate. Weight decay is shared across all groups.

Experiments	Encoder LR	IDM LR	FWM LR	Agg. LR	Text LR	Weight decay
Ours ViT-L and action token dimension ablation	3×10^{-5}	3×10^{-5}	3×10^{-5}	3×10^{-5}	5×10^{-6}	0.01
Ours DINOv2 and action token dimension ablation	2×10^{-5}	1×10^{-4}	5×10^{-5}	1×10^{-4}	5×10^{-6}	0.01
Ours SurgeNet and action token dimension ablation	5×10^{-5}	5×10^{-5}	5×10^{-5}	1×10^{-4}	5×10^{-6}	0.01
No IDM&FWM ViT-L Baseline	3×10^{-5}	-	-	3×10^{-5}	5×10^{-6}	0.01
No IDM&FWM DINOv2 Baseline	2×10^{-5}	-	-	1×10^{-4}	5×10^{-6}	0.01
No IDM&FWM SurgeNet Baseline	5×10^{-5}	-	-	1×10^{-4}	5×10^{-6}	0.01
All other baselines	5×10^{-5}	-	-	-	5×10^{-5}	0.01

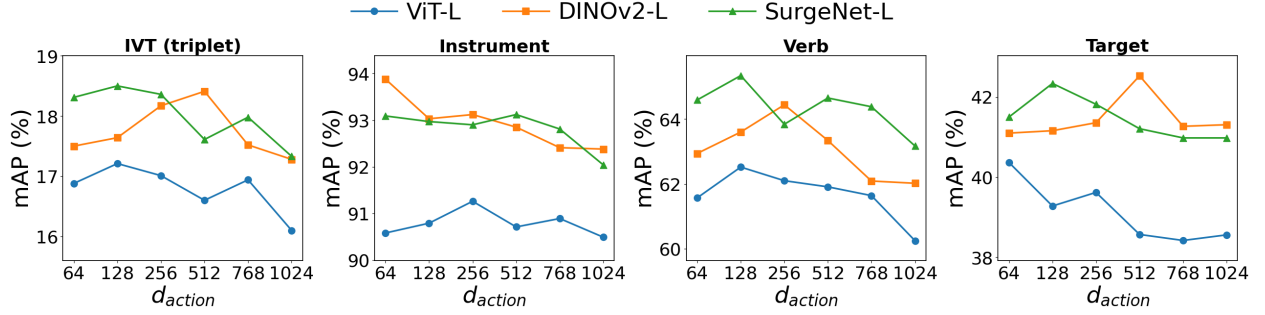


Figure 5: Effect of the action-token dimension d_a on CholecT50. Each panel plots mAP versus $d_a \in \{64, \dots, 1024\}$ for the three encoders (ViT-L, DINOv2, SurgeNet), for the triplet metric and the instrument/verb/target components.

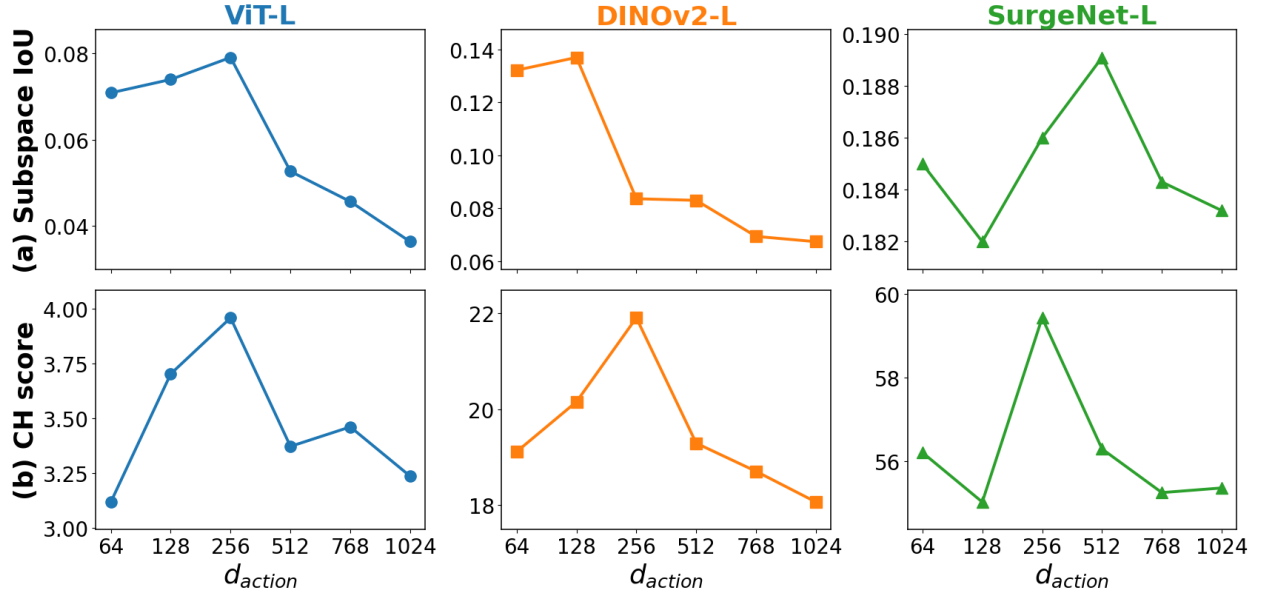


Figure 6: Effect of the action dimension d_a on representation quality for ViT-L, DINOv2-L, and SurgeNet-L. **(a)** Subspace-grounding IoU and **(b)** CH score versus d_a . The grounding IoU is computed from the top- r eigenvectors of the transition covariance with $r=128$ fixed for all models. Each panel uses its own y -scale to make per-encoder trends visible.